Node Depths

AlgoExpert · Easy · Binary Trees

The problem

- Sum of the depths (distance from root) of every node.

Intuition

- Each node contributes its depth, so pass the depth down while traversing and add it up.

Approach

- 1. Traverse from the root at depth zero
- 2. Add the current depth to the total
- 3. Visit children with depth plus one
- 4. Return the sum

Java solution

```java
public class NodeDepths {
    public static int nodeDepths(BinaryTree root) { return sum(root, 0); }

    private static int sum(BinaryTree n, int depth) {
        if (n == null) return 0;
        return depth + sum(n.left, depth + 1) + sum(n.right, depth + 1);
    }
}
```

Complexity

- Time: O(n)
- Space: O(h)

Recap

- Each node contributes its depth, so pass the depth down while traversing and add it up.
- Complexity: O(n)